

SENURI KAVEESHA

Software Engineer | Applied ML & LLM Systems

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PROFILE

Machine Learning Engineer with hands-on experience designing, training, and deploying production-grade ML and LLM-based systems. Strong background in applied ML, NLP, and model optimization, with experience translating complex data problems into scalable solutions. Experienced in end-to-end ML pipelines, model performance optimization, and collaboration with product and engineering teams. Actively working on enterprise AI systems and eager to deepen expertise in finance and tax domain applications.

SKILLS AND COMPETENCIES

Technical Skills

- Core ML & AI : Machine Learning, Applied ML, Model Evaluation, NLP, LLMs, RAG Systems, TensorFlow, PyTorch, scikit-learn, Feature Engineering, Model Optimization
- Data & Analytics : SQL (Complex Queries, Optimization), Data Validation & Accuracy Checks, Pandas, NumPy, Schema Design & Data Pipelines, ETL ELT
- Deployment and Reliability : pilot-to-production AI delivery, Docker, Kubernetes, AWS (ECR), Monitoring, latency optimization, troubleshooting
- Backend & ML Systems : FastAPI, Flask, REST APIs for ML services, Celery, Redis, Model Deployment & Monitoring
- Personal Skills : Analytical Reasoning, Team Collaboration, Quick Adaptation to Change, Precision in Task Management

EXPERIENCE

Work Experiences

Associate Software Engineer - Veracity Digital (Pvt) Ltd

January 2026 - Present

- Designed and developed software systems, including realtime intelligent web apps for real-world applications.
- Built and maintained scalable microservices using Python (backend) and React (frontend), ensuring seamless system integration and performance.
- Implemented AI model development and deployment pipelines, covering training, versioning, testing, and production deployment.

Junior AI Engineer - Parallax Technologies

April 2025 - January 2026

- Partnered with product and engineering teams to deploy production-ready AI solutions integrated into customer-facing applications. Reduced inference latency by 20% by optimizing model serialization.
- Optimized LLM-powered services using Python, TensorFlow, and PyTorch to support real business workflows and product features. Collaborated with product teams to productionize AI models as full-stack features.
- Partnered with product and engineering teams to deploy production-ready AI solutions integrated into customer-facing applications. Translated complex model outputs into actionable insights and deployable system behavior.

Trainee Software Engineer in R&D Department (AI and ML) - CodeGen International (Pvt) Ltd July 2024 - January 2025

Lia Project

- Lowered manual query writing time by 70% by automating SQL generation based on user-provided database.
- Tuned prompt logic, validation layers, and error handling to achieve a 98% query success rate in production-like environments.
- Engineered an automated table structure creation system that maximized query efficiency by 75%.
- Integrated LLM services using OpenAI and LangChain, working closely with engineering and QA teams to ensure reliability and adoption.

Key Technologies:

- Programming Languages and Frameworks: Python (Asyncio, Aiohttp), FastAPI, React js, Next js.
- Database Management: SQL and user-defined database configurations.
- AI/ML Tools: OpenAI and LangChain for LLM-based services, Pinecone, Agent system

- **Lia - QA Automation**

- Increased system reliability and user experience by automating 95% of AI response validations.

Key Technologies:

- Automation Frameworks: Python (Flask, Django), Celery, FastAPI.
- Data Handling and Storage: Redis and SQL databases.

RESEARCH PUBLICATIONS

Adaptive Video Game Content Generation Through Player Centered Modeling (Jan 2024 - April 2025)

- *IEEE International Conference on Advancements in Computing (ICAC), 2025*
 - **IEEE Xplore:** <https://ieeexplore.ieee.org/document/11361432>
- Achieved 91% facial expression recognition accuracy by developing a CNN-based real-time system, delivering adaptive music responses in under 500ms.
- Published research findings at the International Conference on Advancements in Computing (ICAC).
- Resulted in a playable prototype used in user feedback experiments.

PROJECTS

Web application using data mining and machine learning algorithms

(2023 - Group project)

- python, streamlit, GitHub, google colab
- Refined data mining and machine learning algorithms that Streamlined prediction accuracy by 20%.
- Boosted operational efficiency by 15% by leveraging pattern recognition and trend analysis to extract actionable insights.
- Link - https://github.com/IT21925290/FDM_Mini_Project
https://github.com/IT21925290/IRWA_Project

Conversational AI Agent with Continuous Learning

- python, VS code, GitHub, AI
- Initiated a neural network-based chatbot using NLP enabling adaptive learning from user input to improve response accuracy by 30%.
- Engineered a user feedback mechanism that enhanced model performance over time, increasing response accuracy by 30%. The solution utilized NLP techniques in Python for adaptive learning.
- Link - https://github.com/SenuriKaveesha/chatbot_AI
<https://github.com/SenuriKaveesha/Chatbot2>

Cloud native monitoring application

- python, Flask, psutil, AWS(ECR, KBS), kubernetes, docker
- Built a real-time system monitoring tool using PSUtil and Python (Flask), allowing users to track server performance metrics (CPU, Memory, Disk) with a low latency refresh rate of <2 seconds.
- Link - https://github.com/SenuriKaveesha/monitoring_App

Garbage Management System

- **Garbage management system for local council - Balangoda** (2023 - Group project)
- HTML5, CSS3, MongoDB, NodeJS, Express, React, Git, GitHub, MySQL, NextJS
- Link - https://github.com/SLIITIP/y2_s2_wd_it_01-itp_wd_b01_g_09

QUALIFICATIONS

Education

BSc (Hons) in Information Technology - Specializing in Data Science
Sri Lanka Institute of Information Technology (SLIIT) (2021 - 2025)

Certifications

- **Certified in Programming for Everybody (Getting Started with Python) University of Michigan**
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate certification**

EXTRA CURRICULAR ACTIVITIES

IEEE Student Member, SLIIT

Leo Club at SLIIT